

Arsh Kalra

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EDUCATION

Vellore Institute of Technology, Chennai (CGPA : 8.5)

Aug 2023 - May 2027

B.Tech in Computer Science. Coursework includes Data Structures & Algorithms, Operating Systems, Database Management Systems, Web Technologies, Machine Learning, and Object-Oriented Programming.

Amity International School , New Delhi

July 2009 - Apr 2023

Central Board of Secondary Education (Class X - 2021 | 98% && Class XII | 94%)

TECHNICAL SKILLS

Languages: Python, C++, C, Java, JavaScript **Frontend:** HTML5, CSS3, React.js, responsive design, component-based architecture. **Backend & APIs:** REST APIs, JSON, basic backend integration. **Machine Learning:** Face recognition systems, model training, preprocessing, feature extraction. **Core CS:** Data Structures & Algorithms, OOPs, DBMS, Operating Systems. **Tools:** Git, GitHub, Linux, VS Code.

WORK EXPERIENCE

Software Engineering Intern | GTROPY (MapmyIndia)

May 2026 - July 2026

- Developed backend services using Python, FastAPI, MongoDB, and Redis to process and analyze real-time GPS telemetry data from large-scale vehicle fleets.
- Built analytics solutions for vehicle stoppage detection, route intelligence, and connectivity/dead-zone analysis, generating operational insights for enterprise logistics customers.
- Contributed to ML-driven logistics optimization, including unloading location prediction and geospatial analytics, while deploying scalable services using Docker and REST APIs.

PROJECTS

FleetInsight - AI-Powered Fleet Telemetry & Logistics Analytics Platform

Developed a real-time fleet intelligence platform that processes GPS telemetry data to generate actionable logistics insights. Built analytics modules for vehicle stoppage detection, route intelligence, connectivity dead-zone analysis, and unloading point prediction using machine learning and geospatial analytics. Leveraged Python, FastAPI, MongoDB, Redis, and Docker to design scalable backend services and data pipelines for enterprise fleet operations.

StructScan - AI-Based Infrastructure Damage Detection System

Developed an AI-based infrastructure monitoring system to detect structural cracks using a YOLO-based computer vision model. The system assesses crack severity, classifies structural risk levels, and integrates real-time monitoring using an ESP32 camera for surveillance applications.

AgroScan - AI Powered Crop Disease Detection and Plant Health Assessment System

Built a web-based crop health assessment platform that detects plant diseases from leaf images and classifies plant health conditions. Implemented image processing workflows, automated disease detection, and real-time diagnostic reporting with treatment suggestions. Focused on creating an accessible system that helps users quickly analyze crop health through a simple image-based interface.

SafeStreets - AI-Based Violence Detection System

Developed an AI-powered violence detection system using computer vision and deep learning to analyze video streams and identify violent activities in real time, supporting automated surveillance and enhanced public safety.

Intelligent Ride Booking Application

Designed and developed an intelligent ride booking application that simulates real-world ride-hailing workflows. The system uses rule-based logic for ride matching, distance-based fare estimation, and real-time ride status updates, with a focus on usability and software design.

CERTIFICATIONS

Completed industry-relevant certifications in web development, data structures, and artificial intelligence, including the Machine Learning Specialization by Andrew Ng , Meta Front-End Developer Certification, and JavaScript Algorithms & Data Structures. Additionally, gained foundational knowledge in data analytics and cloud technologies through certifications such as Google Data Analytics and AWS Cloud Practitioner, strengthening skills in problem-solving, full-stack development, and AI-driven applications.